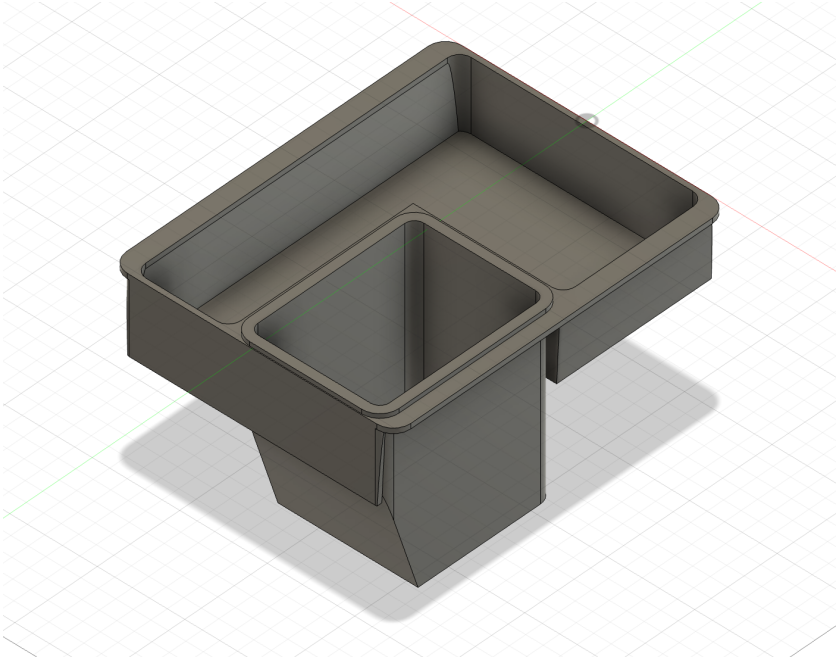
The baseline and the brief.

A tapered console tray and removable rear-passenger bin for the 2024 Tesla Model Y context. This is an independent design project, not a Tesla product or a verified fit certificate.



Keep the accepted base. Design the lid.

The tray and bin are frozen at revision 1.3.0. Propose changes in an item-specific review thread before modifying the baseline. Comments do not change the accepted CAD.

Four millimeters above the flange.

The flange top is Z = +2.5 mm. Chris confirmed 4 mm of additional headroom, so the closed/sliding ceiling is Z = +6.5 mm relative to the carrier seating plane. Hinge hardware, tolerances, warping, flex, seals, and running clearance all share this allowance.

Open downward into the bin.

No finished lid, hinge axis, or validated swept volume is supplied. Check the entire mechanism against the walls, optional liner, accumulated trash, lift-out path, and console cover. Manufacturing process, materials, tolerances, and mechanism choice belong to the designer.

The dimension register.

All values are in millimeters unless marked in degrees. Rounded nominal openings exclude corner reductions. The website provides an anchor and explanation for every item.

FEATURE	VALUE	EVIDENCE
Overall rail width	**162.5 mm**	CAD verified
Overall length	**203 mm**	CAD verified
Tray outside depth	**44.5 mm**	CAD specification
Body width beneath rails	**146.5 mm**	CAD verified
Body width at bottom	**142.5 mm**	CAD verified
Rail plate thickness	**3 mm**	CAD specification
Rail projection	**8 mm**	Derived
Tray side wall	**3 mm**	CAD verified
Tray floor	**3 mm**	CAD specification
Usable tray depth	**41.5 mm**	Derived
Front cavity width	**140.5 mm**	Derived
Front cavity length	**85.5 mm**	Derived
Side cavity at top	**54.355 mm**	Derived
Side cavity at floor	**52.5 mm**	Derived
Receiver reinforcing rib	**6 mm**	CAD specification
Minimum rim support	**2.5 mm**	CAD verified
Receiver clearance per side	**0.5 mm**	CAD specification
Receiver rear wall	**3 mm**	Derived
Bin upper width	**80 mm**	CAD verified
Bin upper length	**105 mm**	CAD verified
Bin maximum depth	**105 mm**	CAD verified
Bin overall height	**107.5 mm**	Derived
Bin shell thickness	**2.4 mm**	CAD specification
Bin floor thickness	**3 mm**	CAD verified

Dimensions / continued.

All values are in millimeters unless marked in degrees. Rounded nominal openings exclude corner reductions. The website provides an anchor and explanation for every item.

FEATURE	VALUE	EVIDENCE
Original outside corner radius	**8 mm**	CAD specification
Flange outside width	**86 mm**	CAD verified
Flange outside length	**111 mm**	CAD verified
Flange above seating plane	**2.5 mm**	CAD verified
Nominal opening width	**75.2 mm**	Derived
Nominal opening length	**100.2 mm**	Derived
Opening corner radius	**5.6 mm**	Derived
Lid height allowance	**4 mm**	User confirmed
Closed/sliding height limit	**6.5 mm**	Derived
Rear relief begins	**44.5 mm**	CAD verified
Rear forward setback	**3 mm**	CAD verified
Rear wall from vertical	**18°**	CAD verified
Sloped section height	**60.5 mm**	Derived
Bottom outside length	**82.342 mm**	CAD verified
Total rear-bottom inset	**22.658 mm**	Derived
Inside shoulder depth	**42.1 mm**	Derived
Sloped rear wall thickness	**2.4 mm**	CAD verified
Reference slide travel	**77 mm**	Unverified reference
Bin inside shallow envelope	**1.145 mm**	CAD verified

What the evidence supports.

Verified in Fusion

Two connected solids. Zero assembly interference. The accepted source passed 29 regression tests and 53 point-containment checks. Exported part volumes match the approved validation report. The browser mesh is tessellated from the same BReps at a requested 0.02 mm surface tolerance.

01 - Carrier With L-Shaped Tray And Open Passenger Side: 190615.223 cubic millimeters. Bounds: -81.25, 81.25, 0, 203, -44.5, 0.

02 - Removable Rear Passenger Trash Bin: 108378.631 cubic millimeters. Bounds: -73.105, 12.895, 91.5, 202.5, -105, 2.5.

Unverified or conflicting

Carrier width 162.5 exceeds original reference 157.0 mm.

Body width 146.5 exceeds original reference 143.5 mm.

Bin exceeds centered 130.0 mm deep-width reference by 5.11 mm on the passenger side; outward shift explicitly approved.

Open-side rim stiffness and print strength require physical testing.

Front reference floor gap: 17.0 mm; old target: 55.0 mm.

Rear reference floor gap: 50.0 mm; old target: 55.0 mm.

Rear relief is 18 degrees from vertical with a 3 mm forward step; this does not establish a measured 3 mm console gap.

Physical fit, lift-out, and the full 77 mm slide are unverified.

Coordinate system

X: Positive: driver. Negative: passenger.

Y: Positive: rear. Zero: vehicle front.

Z: Zero: carrier seating plane. Positive: upward.

Files and provenance

Assembly and individual-part STEP files are exact CAD exports. The F3D is the approved native archive. This is a direct-design BRep model, not a fully parametric feature-history design. A SHA-256 manifest identifies the downloadable assets. No manufacturing tolerances or vehicle compatibility are implied.

Generator SHA-256:

ad4fe266e3c240667cce263d63aad974f5d9cb6f6659b6c4f52c72362b6b7f5c

Review: https://cichonship.github.io/tesla-console-review/